BIOLOGY 121

(Genetics, Evolution & Botany)

Mendelian genetics

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Introduction

- Genetics is a sub-field of biology.
- Biology is a sub-field of science dealing with ;
  - **Epistemology**: investigates origin, nature, methods and limits of human knowledge
  - **Metaphysical**: concerned with abstract thought or subject as existence, causality or truth
  - **Ethical**: pertaining morals or principles of morality, issues of right or wrong in conduct

Introduction

- Biology considers the philosophical questions raised by nature and existence of living organisms and biological processes.
  - What is the origin of life?
  - Why are their variations among parents and offspring?
  - What is the nature of evolutionary theory?

  etc

Theories of the origin of life

- **Spontaneous Generation** states that living things arise from non living things (water+light+dust+heat).
- Spontaneous generation was disproved by Louis Pastuer who proposed the **Cell theory**.

  a) Every cell comes from another cell, b) the cell is the basic unit of life
- The cell theory fails to explain: -

  a) variations that appear among parents and offspring, b) the virus which is a living thing but a cellular c) the origin of the first cell

Theories of the origin of life

- Lamarck, proposed a theory of **Inheritance of Acquired Characteristics**. This states that changes in the environment create new needs which led to new behaviours that may involve use or disuse of some structures.
- The theory of inheritance of acquired characteristics was disproved by Wiseman who cut tails of the rats expecting that the offspring will have no tails.

Theories of the origin of life

- Wiseman proposed the **Blending theory** which stated that characteristics of offspring are as a result of combination of traits of both parents which blend together.
- The shortfall of this theory is that it does not explain the presence of some characteristic that are not present in either of the parents e.g. albino offspring from normal parents.

Theories of the origin of life

- Mendel came up with his theory called **Inheritance theory** which states that similarities and differences between parents and off springs are as a result of the passage of discrete hereditary factors (genes). It is also referred as Particulate theory.

Theories of the origin of life

- Factors that led to the success of Mendel are:-

  1. Knowledge of statistics which he used to predict and verify outcome of his experiments
  2. Choice of subjects for experiments, he used garden peas which can be cross pollinated or self pollinated
  3. Choice of characters, he looked at one character at a time
  4. The way of presenting results, he used numbers to represent whatever he discovered
  5. His background, he was interested in breeding experiments

Mendelian Genetics

- Mendel’s monohybrid and dihybrid experiments gave the idea of dominance and recessive where the recessive gene is masked by the dominant gene and the recessive gene can be expressed only in homozygous state (dd).
- From his experiments he also came up with 2 important laws:-
  - Mendel’s law of segregation:- states that genes segregate during gamete formation and reunite randomly during fertilisation.
  - Mendel’s law of independent assortment:- states that inheritance of a gene pair on a chromosome is unaffected by inheritance of other gene pairs on the other chromosome.

Monohybrid cross

- **Monohybrid cross**: a genetic cross involving a single pair of genes (one trait)
- Used by geneticists to observe how **homozygous** offspring express h**eterozygous** genotypes inherited from their parents.
- **P** = Parental generation
- $\mathbf{F_1}$ = First filial generation
- $\mathbf{F_2}$ = Second filial generation of a genetic cross

Mendelian ratios

A. Monohybrid inheritance

Mendel took pure lines that breed true

Parents	DD (tall)	x	dd	(dwarf)
Gametes	DD		dd	(homogametic)
F1	Dd (all tall)			

He then took Dd and self-pollinated (selfed)

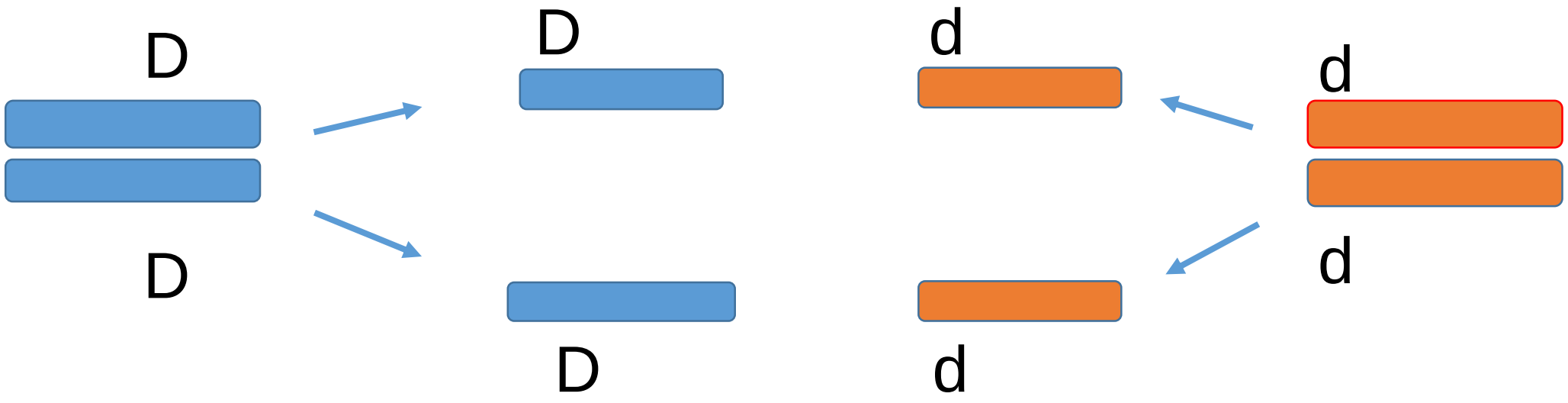
Parents (F1)	Dd (tall)	x	Dd (tall)	
Gametes	Dd		Dd	(heterogametic)
F2	DD (tall)	Dd (tall)	Dd (tall)	dd (dwarf)

This gave a 3:1 tall to dwarf (this is a Mendelian monohybrid ratio)

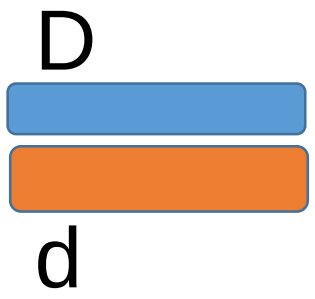
Parent 1 (tall)
Homogametic
Homozygous

Parent 1 (dwarf)
Homogametic
Homozygous

Gametes



First filial generation (F1)
Heterozygous



Punnett square

- A useful tool to do genetic crosses
- It is used to predict the genotypes and phenotypes of the offspring.
- For a monohybrid cross, the square is divided into four
- Looks like a window pane

Using a Punnett Square

STEPS:

1. determine the genotypes and gametes for each parent organisms
2. write down your "cross" (mating)
3. draw a p-square

E.g Parent genotypes :TT and t t

Gamates : T ,T X t, t

Punnett square

4. **"split" the letters of the genotype for each parent & put them "outside" the p-square**

5. determine the possible genotypes of the offspring by filling in the p-square

6. summarize results (genotypes & phenotypes of offspring)

T T × ***t t***

	T	**T**
t	**T *t***	**T *t***
t	**T *t***	**T *t***

Genotypes:
100% T t

Phenotypes:
100% Tall plants

Monohybrid cross: F_2 generation

- If you let the F1 generation self-fertilize, the next monohybrid cross would be:

$$\mathbf{T}\,\boldsymbol{t} \times \mathbf{T}\,\boldsymbol{t}$$

(tall) (tall)

	T	***t***
T	**T T**	**T *t***
t	**T *t***	***t t***

Genotypes:
1 TT= Tall
2 T*t* = Tall
1 *tt* = dwarf
Genotypic ratio= 1:2:1

Phenotype:
3 Tall
1 dwarf

Phenotypic ratio= 3:1

Dihybrid crosses

- A dihybrid cross is a type of genetic cross between two individuals with either homozygous or heterozygous genotypes of two characters or traits.

- The dihybrid cross is used as a method to determine or prove the law of independent assortment as it proves that two characters segregate independently of one another.

- It only works in the case of genes that are not linked and are present on different chromosomes.

Mendelian ratios

B. Dihybrid inheritance

Mendel took peas with round (W) and green (Y) seeds and crossed them with wrinkled (w)and yellow (y) seeds.

Parents WWYY x wwyy

↓ ↓

Gametes WYWY wywy

Gametes	WY	WY
wy	WwYy	WwYy
wy	WwYy	WwYy

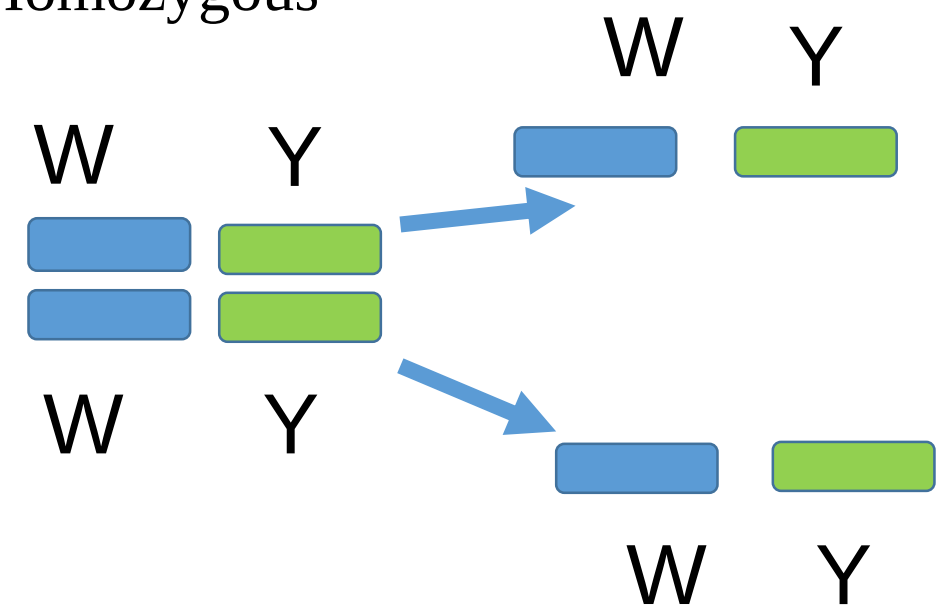
F1 : WwYy (round and green seeds)

He then self-fertilization F1 (for this we use a Punnett square)

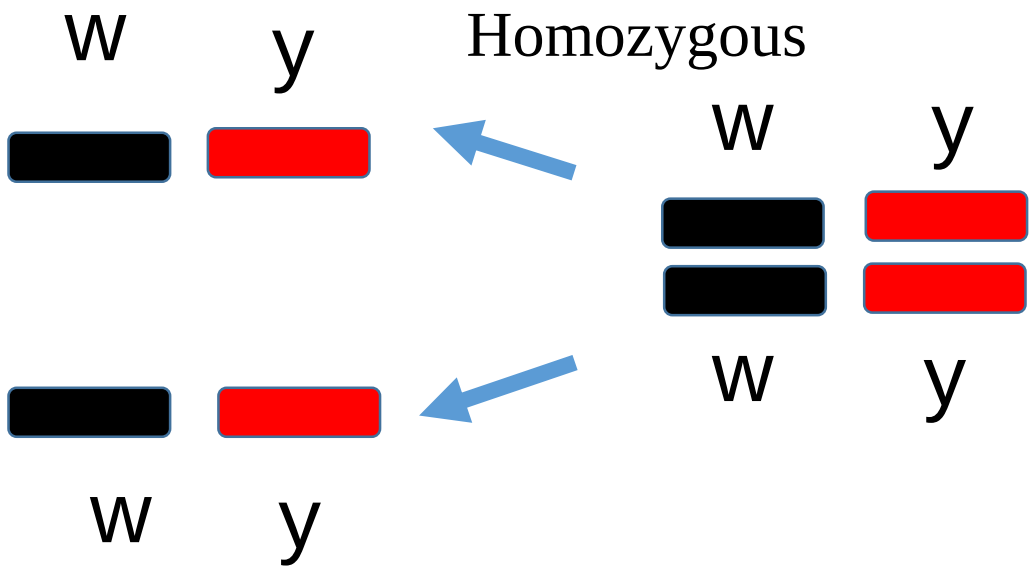
Gametes	WY	Wy	wY	wy
WY	WWYY rg	WWYy rg	WwYY rg	WwYy rg
Wy	WWYy rg	WWyy ry	WwYy rg	Wwyy ry
wY	WwYY rg	WwYy rg	wwYY wg	wwYy wg
wy	WwYy rg	Wwyy ry	wwYy wg	wwyy wy

- This gave a Mendelian dihybrid ratio of 9:3:3:1 of
- round green seeds: round yellow seeds: wrinkled green seeds: wrinkled yellow seeds

Parent 1 (round green seeds)
Homogametic
Homozygous

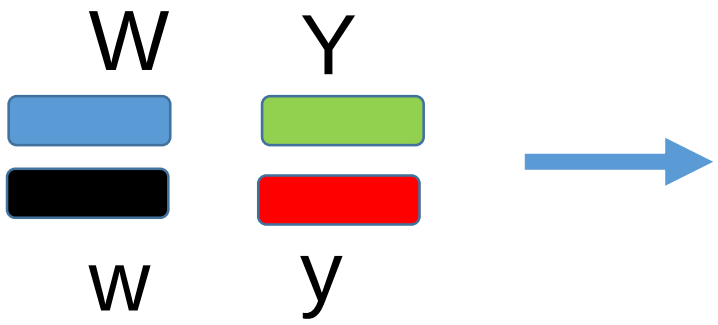


Gametes



Parent 1 (wrinkled yellow seeds)
Homogametic
Homozygous

First filial generation (F1)
Heterozygous
WwYy



Gametes would be
WY, Wy, wY or wy

Test cross

We saw that both DD and Dd had tall phenotypes while WWYY and WwYy also had dominant phenotypes (round green seeds).

A test cross is conducted to determine whether an individual is either homozygous or heterozygous?

Parents	DD x	dd
Gametes	D	d
F1	Dd	(all tall)

Parents	Dd	x	dd	
Gametes	D d		d	d
F1	Dd	Dd	dd	dd

Dd and dd (50% tall and 50% Dwarf)

Example for Monohybrid cross

1. A **homozygous** lion with blue eyes **(bb)** mates with a **heterozygous** lion with brown eyes **(Bb)**

a. Calculate the genotype and phenotype ratios?

b. What is the probability that will produce a lion with blue eyes?

Gametes : bb and Bb

	b	**b**
B	**Bb**	**Bb**
b	**bb**	**bb**

Genotype ratios

Bb : bb

2: 2

1:1

Phenotype

Blue : Brown

2 : 2

1:1

P = 2/4=0.5*100 =50%

Example for Dihybrid cross

In cats, the ability to run normally is a dominant trait. A cat with these traits is called running cat (R). The recessive trait causes cats to run in circles only. Cats with these traits are called waltzing cats (r). The hair color is also inherited in cats. Black hair(B) is dominant over brown hair b.

- Using a Punnett square cross a **heterozygous running**, **heterozygous black** cat **with homozygous running, homozygous black cat**

Example for Dihybrid cross

In cats, the ability to run normally is a dominant trait. A cat with these traits is called running cat (R). The recessive trait causes cats to run in circles only. Cats with these traits are called waltzing cats (r). The hair color is also inherited in cats. Black hair(B) is dominant over brown hair b.

Using a Punnett square cross a heterozygous running, heterozygous black cat with homozygous running, homozygous black cat

	Parent 1		**Parent 2**
Parent phenotypes:	Heterozygous running & black		Homozygous running & black
Parent Genotypes:	RrBb	x	RRBB
Gametes:	**RB,Rb,rB,rb**		**RB, RB, RB, RB**

Gametes	RB	Rb	rB	rb
RB	RRBB	RRBb	RrBB	RRBb
RB	RRBB	RRBb	RrBB	RRBb
RB	RRBB	RRBb	RrBB	RRBb
RB	RRBB	RRBb	RrBB	RRBb

Off Spring phenotypes % : Running and black = 100%

Running and brown = 0%

Waltzing and black = 0%

Waltzing and brown = 0%

2. In mouse, brown coloring (B) is dominant, and blue coloring (b) is recessive. A homozygous brown mouse crosses with a heterozygous mouse. Calculate the genotype and phenotype ratios?

3. A pure-breeding pea plant with round seeds and white flowers is crossed with a pure-breeding plant that has wrinkled seeds and purple flowers.

i. what will the F1 offspring look like?

ii. if two F1 plants are crossed, what phenotypes will be expected in the F2 generation, and in what proportions?

3. In a dihybrid cross, AaBb x AaBb, what fraction of the offspring will be homozygous for both recessive
traits?

4. Following a SSYy x SsYy cross, what fraction of the offspring are predicted to have a genotype that is heterozygous for both characteristics?

5. Wolves are sometimes observed to have black coats and blue eyes. Assume further that normal coat color (N) is dominant to black (n) and brown eyes (B) are dominant to blue (b). Suppose the alpha male and alpha female of a pack (these are the dominant individuals who do most of the breeding) are black with blue eyes and normal colored with brown eyes, respectively. The female is also heterozygous for both traits.

What are the possible genotypes of the offspring?

Practice questions

1. In guinea pigs, rough coat R is dominant over smooth coat r, and black coat B is dominant over white b. R and B are independent genes (not linked). If a rough black guinea pig is bred with a rough white and gives 28 rough black; 31 rough white; 11 smooth black and 10 smooth white.

What are the likely genotypes of the following offspring?

Rough black

Rough white

Smooth black

Smooth white

What are the likely genotypes of the parents?

Modification of Mendelian ratios

1. Incomplete dominance
2. Codominance
3. Multiple alleles/genes
4. Lethal genes
5. Duplicate genes
6. Epistasis
7. Non-epistasis
8. Complementary
9. Polygenic
10. Linkage genes
11. Mutations

Modification of Mendelian ratios

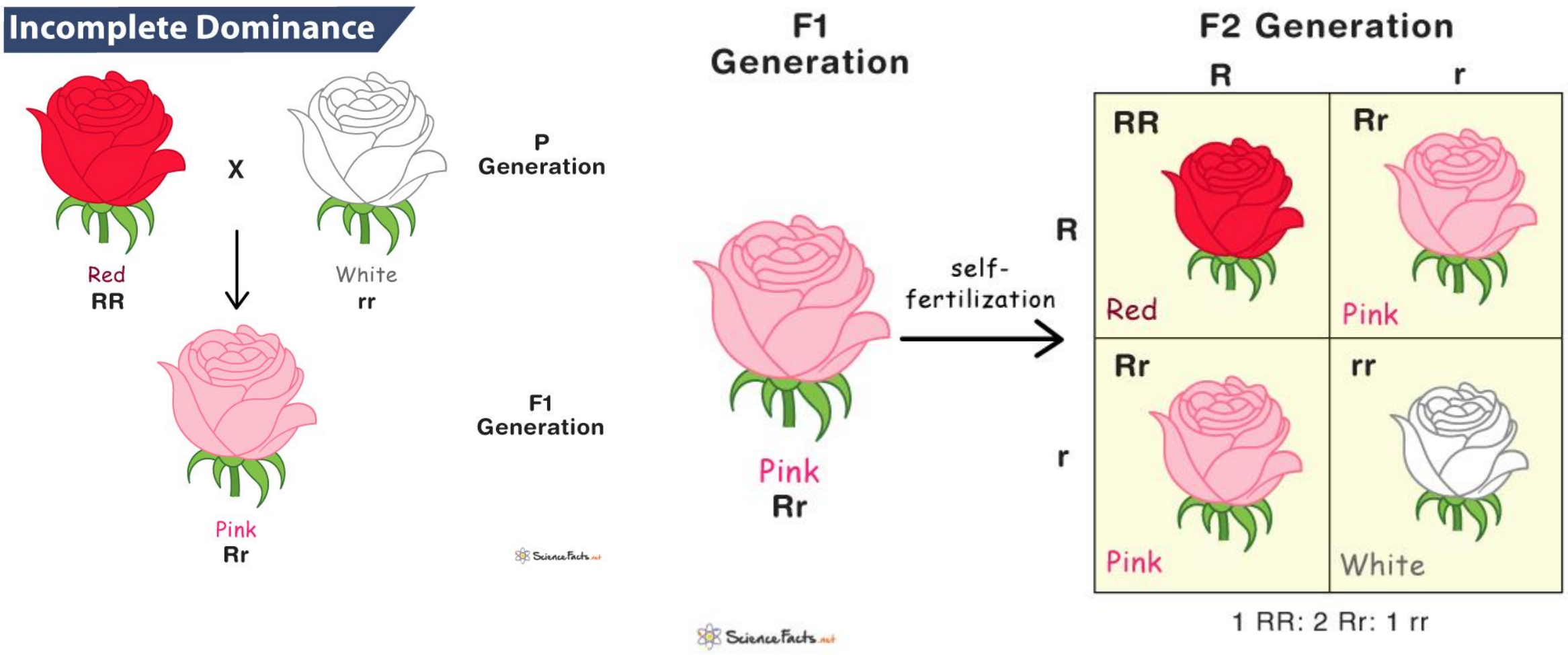
1. Incomplete dominance

Products of both genes/alleles give an intermediate character because non is completely dominant over the other.

e.g. Flower colour:

Parents1:	**RR (red)**	**x**	**rr (white)**
Gametes:	RR		rr
F1	**Rr (pink)**		
Parents 2:	**Rr (pink)**	x	**Rr (pink)**
Gametes:	R , r		R, r

F2 **RR Rr Rr rr (ratio of 1 : 2 : 1 of red : pink : white)**



Recollect that the genotype ratio of F2 generation in the monohybrid cross by Mendel also gave the same ratio of 1:2:1. However, the phenotype ratio has changed from **3:1 to 1:2:1.** The reason for this variation is the incomplete dominance of the allele R over the allele W. This led to the blending of color in flowers

Modification of Mendelian ratios

2 **Codominance**

Both genes/alleles are expressed in the offspring like MN blood group

e.g. MN blood group:

Parents 1: **L^ML^M (M) x L^NL^N (N)**

Gametes: L^M , L^M L^N,L^N

F1 **L^ML^N (MN)**

Parents 2: **L^ML^N (MN)** x **L^ML^N (MN)**

Gametes L^M L^N L^M L^N

F2 **L^ML^M L^ML^N L^ML^N L^NL^N**

Phenotype: **1: 2:1**

; Genotype: **1:2:1**

Modification of Mendelian ratios

- **Multiple genes/alleles**
- This occurs when there is more than one form of alleles in individuals but only two forms of alleles can exist in a population.
- The AOB blood group system is an example of a multiple allele system, meaning that there are more than two possible alleles that can determine an individuals blood type within the system.
- In the AOB system, there are three main alleles: A,O and These alleles interact in various combination to determine an individual’s blood type

- There are multiple possible genotypes and phenotypes within the AOB blood group system

Genotype	Blood group
$I^A I^A$	A
$I^A I^B$	AB
$I^A I^O$	A
$I^B I^B$	B
$I^B I^O$	B
$I^O I^O$	O

Examples

1.What will the **F1 generation's** blood group be if you cross a Mother with blood **group O** with a Father with **blood group AB**?

Parent	**Phenotype**	**Father Group AB**	**Mother Group O**
(p)	Genotype	$I^A I^B$	$I^O I^O$
	Gametes	I^A I^B	I^O I^O

Multiple genes

Male

Female

	I^A	I^B
I^O	$I^A I^O$	$I^B I^O$
I^O	$I^A I^O$	$I^B I^O$

Offspring genotype: 50% $I^A I^O$,50% $I^B I^O$

Offspring Phenotype: 50% **blood group A**

50% **blood group B**

2. If a father with **blood group A (heterozygous)** has children with a mother with **blood group B (Heterozygous)**, what are the possible blood groups of the children?

Parent	**Phenotype**	**Father Group A**	**Mother Group B**
(p)	Genotype	$I^A I^O$	$I^B I^O$
	Gametes	I^A I^O	I^B I^O

Male

Female

	I^A	I^O
I^B	$I^A I^B$	$I^B I^O$
I^O	$I^A I^O$	$I^O I^O$

Offspring genotype: 25% $I^A I^B$,25% $I^B I^0$

25% $I^A I^0$,25% $I^O I^O$

Offspring Phenotype: 25% **blood group AB**

25% **blood group B**

25% **blood group A**

25% **blood group O**

Practice questions

1. A person with type AB blood is crossed with another individual. They have 4 childrens and all of them have either type A or type B blood. What are their genotypes.

Show Punnett square to support your answer

2. A woman with type A blood has parents who are both type AB and a husband who is a type B. They have a child with type A blood.

a. What are the genotypes of the woman, her husband and their child?

3.The mother has type A blood. Her husband has type B blood. Their child has type O blood. The father claims the child can’t be his. Is he right?

Modification of Mendelian ratios

4. **Lethal alleles/genes**

- This involves genetic defects that cause a 100% mortality rate.
- It causes death in individuals having a particular pair of alleles to die.
- They are usually a result of mutations in genes that are essential for growth or development.
- The genes are usually recessive and homologous recessive individuals die.
- Could also be dominant or conditional depending on the gene or genes involved

Example: Coat color in mice

A = agouti (brown) = wild-type allele

* A^Y = yellow = mutant allele

P1 **AA agouti × AA agouti**

↓

F_1 **AA Agouti**

AA^Y Yellow × AA^Y Yellow

↓

AA agouti	**AA^Y yellow**
A^YA yellow	A^YA^Y Lethal

↓

Phenotype: 1:2 (1 died)

AA Agouti × AA^Y Yellow

↓

AA agouti	**AA^Y yellow**

↓

Phenotype: 1:1 (All survived)

5. Duplicate genes

- This occurs if a gene is duplicated on the same chromosome. Resulting in multiple copies of the same

e.g. If a gene controlling height is duplicated on a chromosome

Parents 1: AABB (tall) x aabb(short)
Gametes: AB ab
F1: AaBb (tall)
Parents 2 AaBb x AaBb
Gametes: AB/Ab/aB/ab AB/Ab/aB/ab

	AB	Ab	aB	ab
AB	AABB	AABb	AaBB	AaBb
Ab	AABb	Aabb	AaBb	Aabb
aB	AaBB	AaBb	aaBB	aaBb
ab	AaBb	Aabb	aaBb	aabb

All offspring will be tall except one. The ratio will be 15:1 of tall to short.

Modification of Mendelian ratios

6. **Epistasis genes**

This is a situation where there is a gene at locus A that masks the expression of another gene at **locus B**

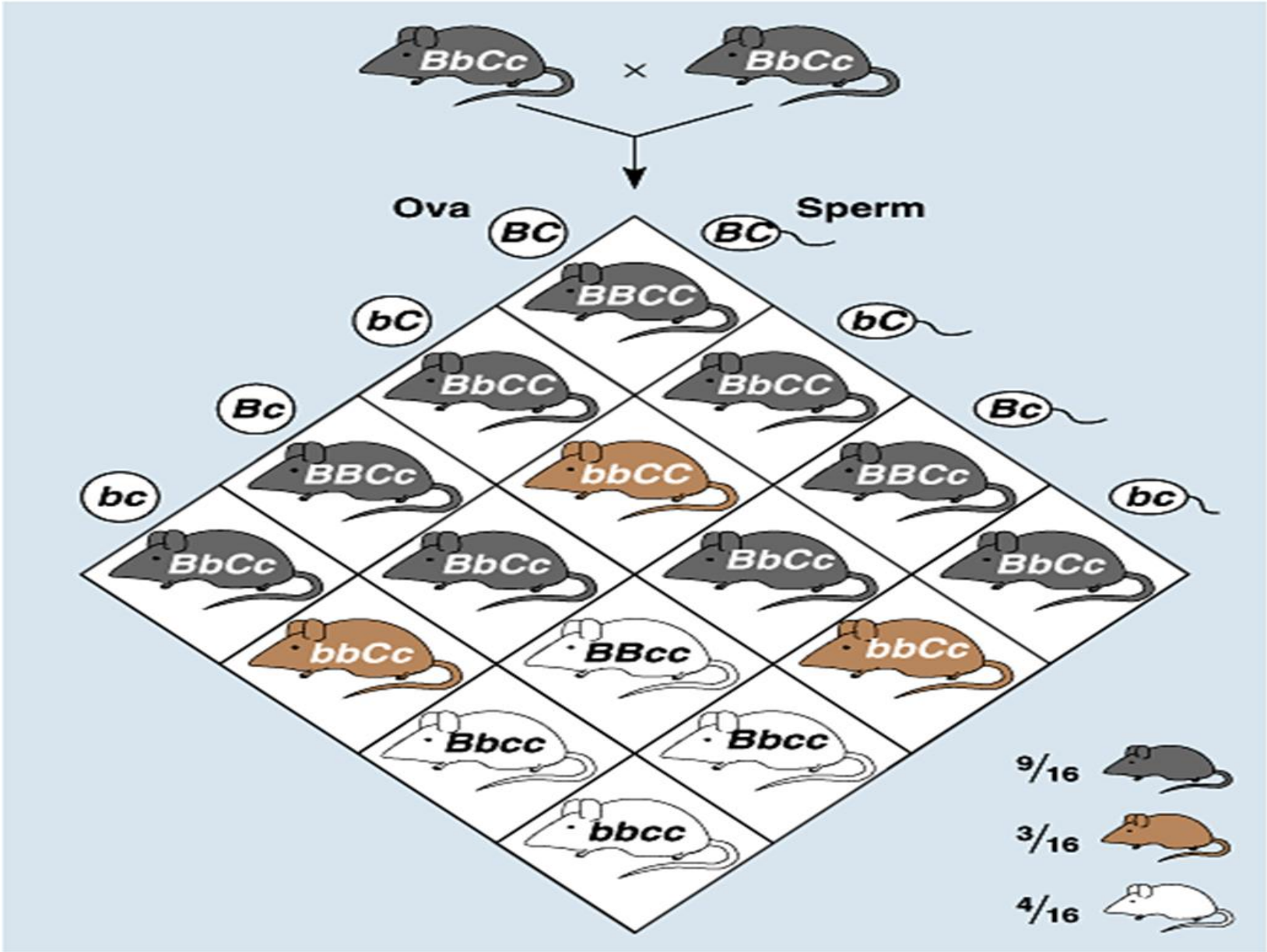
Locus A suppresses the expression of the gene at locus B and it depends on whether there is a **dominant** or a recessive gene on locus B.

Gene at one locus alters the phenotypic expression of a different gene at a second locus i.e. Hair color in mammals: mice

In mice, black (B) coat is dominant to brown coat color (b)

A second gene determines whether or not pigment will be deposited in the hair…dominant (C) for color results in deposition

If the mouse is cc, it will be albino regardless of the genotype at the black/brown locus.



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Modification of Mendelian ratios

7. Non-epistatic genes

- This is a situation where a **character** is a product of a **number of gene products**.
- The products interact to produce a phenotype.
- This inheritance occurs if the end product of a different gene contributes to the same character.

e.g. Flower colour.

Locus A	Locus B	Locus C	Phenotype
AA	BB	CC	Red
Aa	BB	CC	Red
aa	BB	CC	white

In this case the products of these loci contribute towards the same character - flower colour

Modification of Mendelian ratios

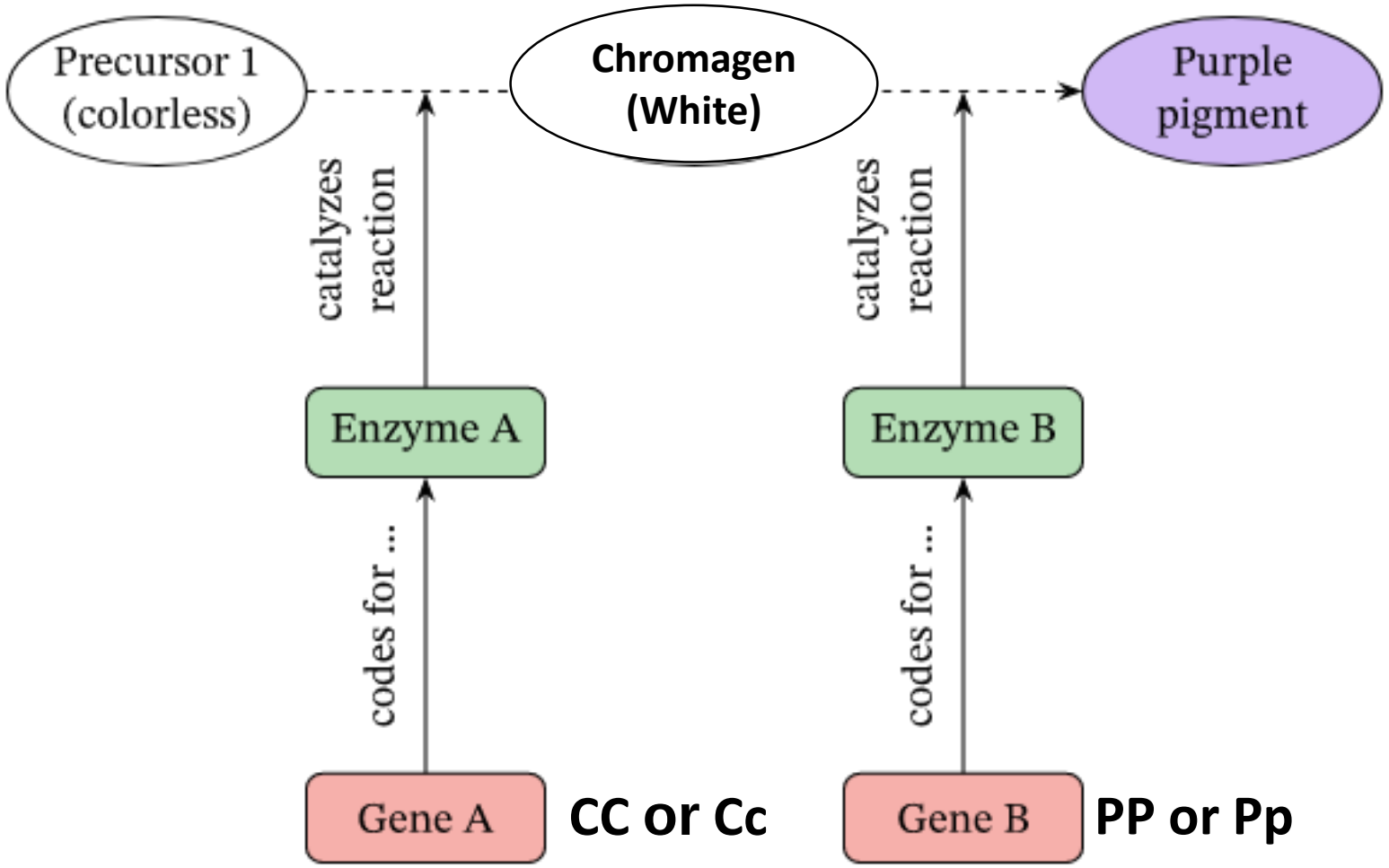
8. Complementary genes

- A type of gene interaction between two different genes that work together to create a specific phenotype or visible trait.
- Both types of genes must be present in **dominant form** to complement each other and produce a specific phenotype

  e.g. if the protein has subunits to be functional (Haemoglobin sub units)

E.g. Colour of flower in *lathyrus* adoratus (sweet pea)

Raw material ……..**Chromagen** (White)……..**Anthocyanin** (purple



Modification of Mendelian ratios

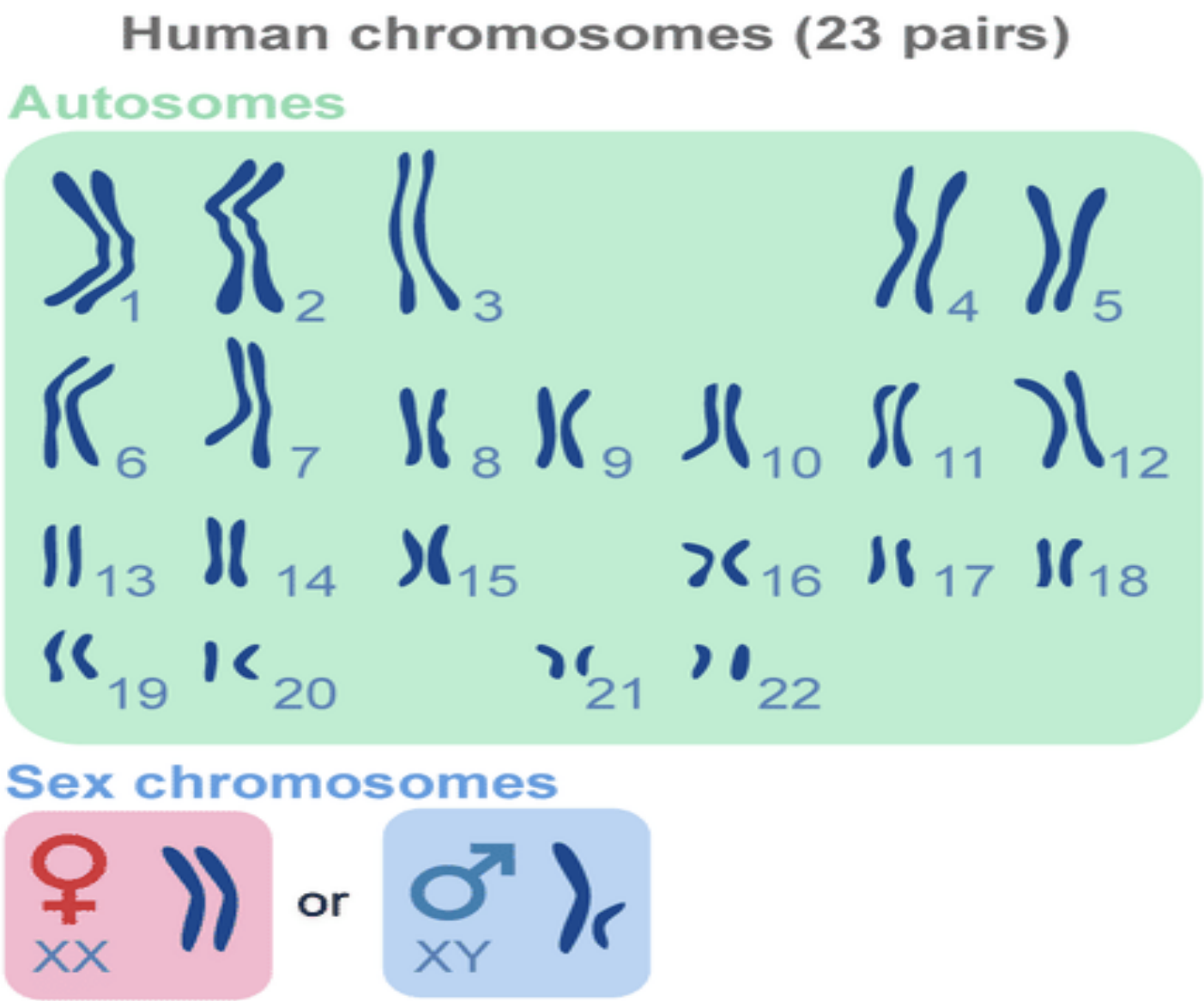
9. Polygenic inheritance

- It is a quantitative inheritance, where multiple independent genes have an additive or similar effect on a single quantitative trait.
- In polygenic inheritance, each allele has a cumulative or additive effect on the phenotype thus generating continuous variation, e.g. skin pigmentation, height, etc.

Modification of Mendelian ratios

10.Linkage

- Genes that are on the same chromosome are called linked genes. These genes are associated in their inheritance, they are inherited together and in the same position.
- The linkages are found either in sex cells/sex chromosomes or in autosomal cells/autosome chromosomes


Human chromosomes (23 pairs)
Autosomes
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
Sex chromosomes
XX
or
XY

Modification of Mendelian ratios

Sex linkage

- These are genes found on the same sex chromosome.
- Mammals have XY in males and females have XX chromosomes.
- Bugs and grasshopper males have only one X chromosome and they are XO, females are XX. There is no Y chromosome
- Birds females have ZW chromosome and males have ZZ
- Examples of sex linked characteristics are 1) colour blindness associated with X chromosome, 2) haemophilia- blood clotting, mainly suffered by females 3) baldness

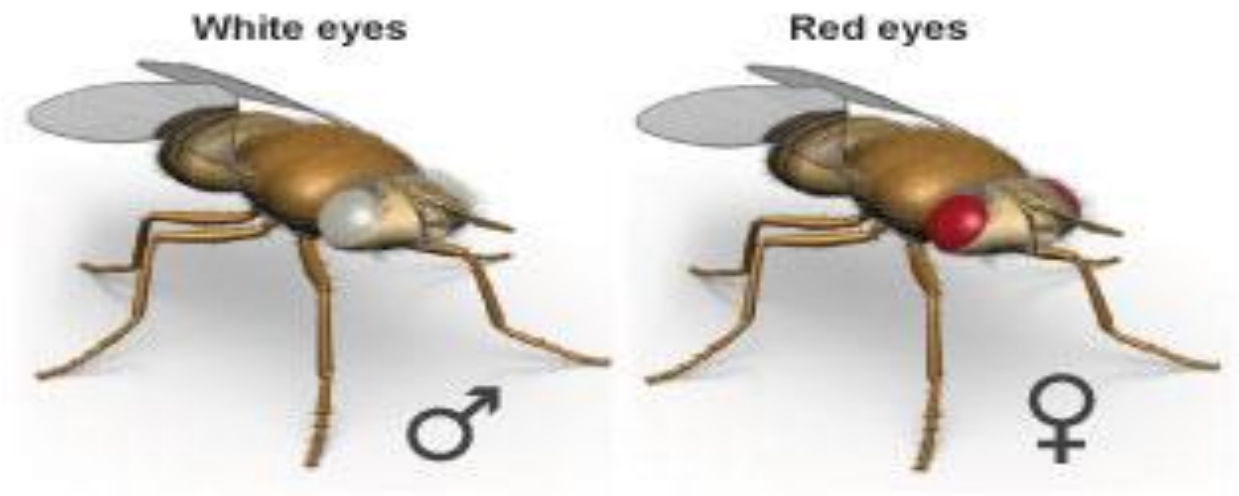
Modification of Mendelian ratios

Sex linkages

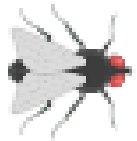
Morgan observed that the inheritance of a character depends on whether the character is from a male or a female parent.

An example - Eye colour of fruit flies :

an experiment done by Dr. Morgan to predict how white eye color is inherited in two generations of fruit flies.



First-Generation Cross



Female fruit fly
Red eye color

Male fruit fly
White eye color

	X^R	X^R
X^r	$X^R X^r$	$X^R X^r$
Y	$X^R Y$	$X^R Y$

Offspring

All (100%) first-generation offspring have a red eye color

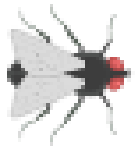
- All female fruit flies carry recessive allele

R = red eye color (dominant allele)

r = white eye color (recessive allele)

Rr = Red eye color

Second-Generation Cross



Female fruit fly
Red eye color

Carrying recessive allele

Male fruit fly
Red eye color

	X^R	X^r
X^R	$X^R X^R$	$X^R X^r$
Y	$X^R Y$	$X^r Y$

Offspring

$X^R X^R$

$X^R X^r$

$X^R Y$

$X^r Y$

In second generation offspring, 75% have a red eye color and 25% have a white eye color

- All white-eyed flies are males

Example

Hemophilia is caused by a mutation in either of two genes, both of which are located on the X chromosome. Both genes encode proteins that help blood clot. Let's focus on just one of these genes, calling the functional allele $\mathbf{X^H}$ and the disease allele X^h

E.g: a woman who is heterozygous for normal and hemophilia alleles($\mathbf{X^H X^h}$) has children with a man who is hemizygous for the normal form**($\mathbf{X^HY}$).** Both parents have normal blood clotting, but the mother is a carrier. What is the chance of their sons and daughters having hemophilia?

What is the chance of their sons and daughters having hemophilia?

Parent	**Phenotype**	**Unaffected father**	**Carrier Mother**
(p)	Genotype	X^HY	$X^H X^h$
	Gametes	X^H Y	X^H X^h

Unaffected father

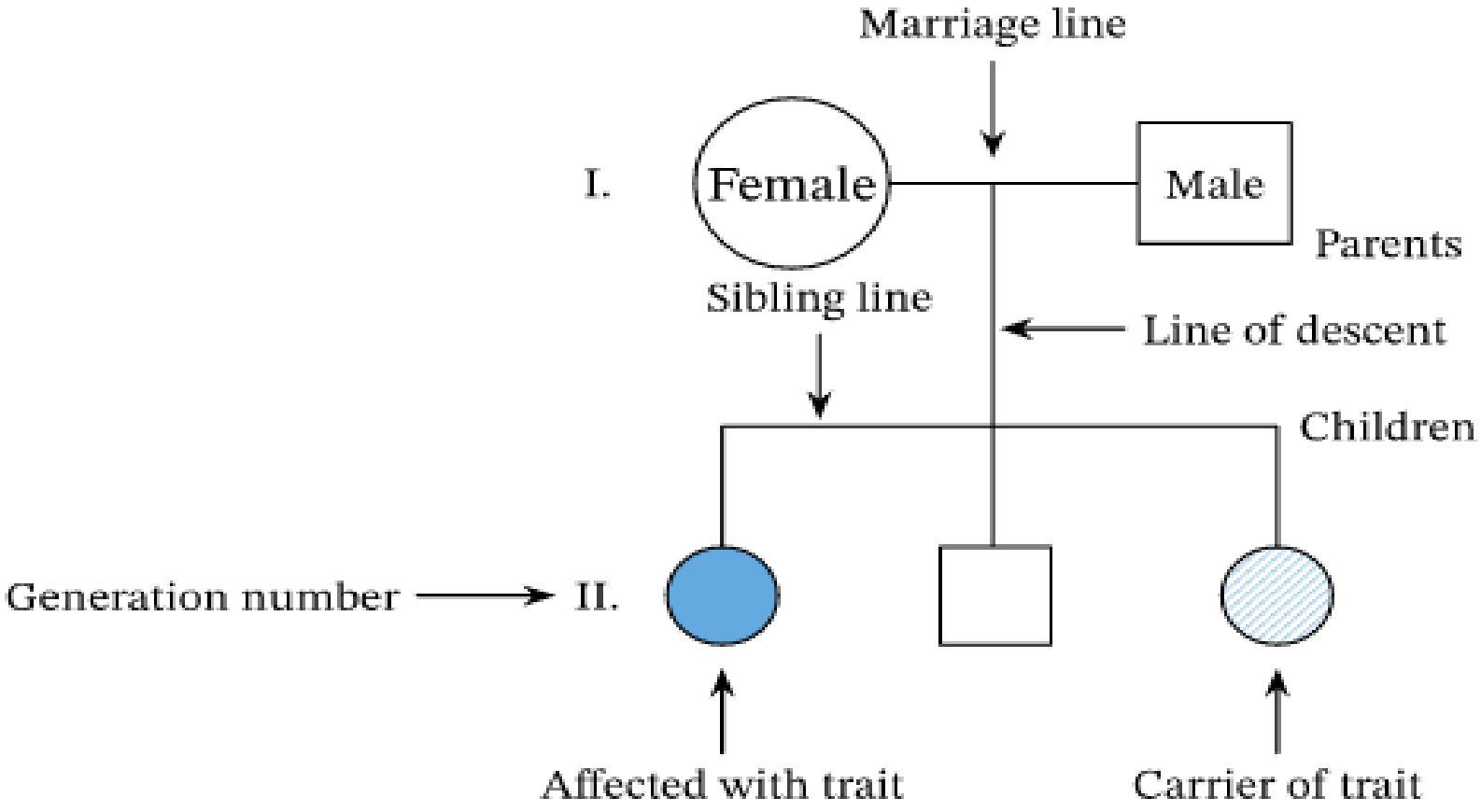
Carrier Mother

	X^H	Y
X^H	X^HX^H	X^HY
X^h	X^HX^h	X^hY

- None of the daughters will have hemophilia
- ½ of the sons will be affected
- ½ of the sons will not be affected

Pedigree Chart

A pedigree chart is a diagram that shows how a gene of a trait or disease is inherited over several generations.



Modification of Mendelian ratios

Autosomal linkage

- There are two types, complete and partial autosomal linkages.
- Genes that are linked completely do not assort themselves independently as a result they are inherited in a block
- Partial linkages come as a result of crossing over at meiosis to make new combinations with other genes and are called recombinants

Modification of Mendelian ratios

- **Autosomal linkage**
  - Frequency of cross over is used to estimate the distance between genes.
    - Genes that are far apart have a higher probability of crossing over than genes that are close together.
    - Genetics have a standard percentage combinations that 1% of recombinants means 1 map unit i.e. The distance between neighbouring genes. 1% = 1 cM (centimorgan)

Modification of Mendelian ratios

Autosomal linkages

e.g. After crossing coloured smooth seeds x coloured wricked seeds the following combinations were found

Coloured smooth seeds 48.25%

Coloured wrickled seeds 48.25%

Colourless wrickled seeds 1.75%

Colourless smooth seeds 1.75%

Total percentage of recombinants for this experiment is 1.75% + 1.75% = 3.5%. Therefore the gene for colour and texture are 3.5cM apart

PRACTICE QUESTIONS

1. Duchenne muscular dystrophy (DMD) is an X-linked recessive condition in humans that causes muscle weakness and wasting. The allele that correctly produces the dystrophin protein (D) is dominant to the allele that causes DMD (d). A female with the genotype $\mathbf{X^DX^d}$ marries a male with the genotype $\mathbf{X^DY}$.

I. What is the probability, in percent, that the offspring will be a male with the disease?

2. The inheritance of color blindness in humans is due to a recessive gene located on the X chromosome (X linked). X^N is normal color sight and is dominant to X^n (color blind)

i. If a color-blind boy is born to parents both of whom have normal vision, what are the genotypes of the three individuals?

ii What is the probability that the couple will have a color-blind daughter?

Modification of Mendelian ratios

11. Mutations

Mutations are sudden changes to the chromosomal DNA that results to some heritable characteristics

Factors that cause mutations also known as mutagens or mutagenic factors have potential of denaturing proteins and DNA are

- X-, alpha-, gamma-, beta- rays
- Ultra violet (UV) light
- Hydrogen peroxide (H_20_2)
- Mustard gas

Modification of Mendelian ratios

Mutations

There are two types of mutation –

a) Gene mutation affecting the gene

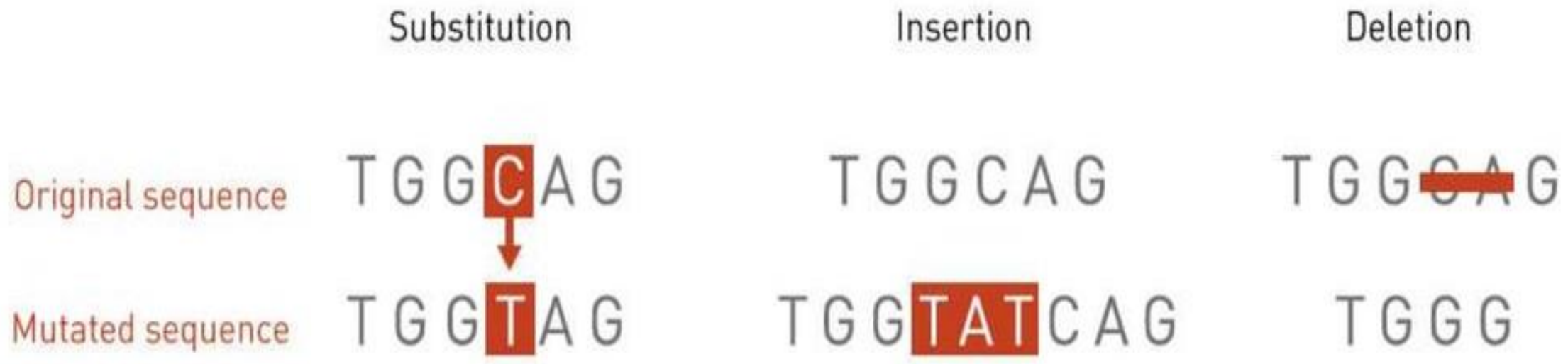
b) Chromosome mutation affecting the chromosome

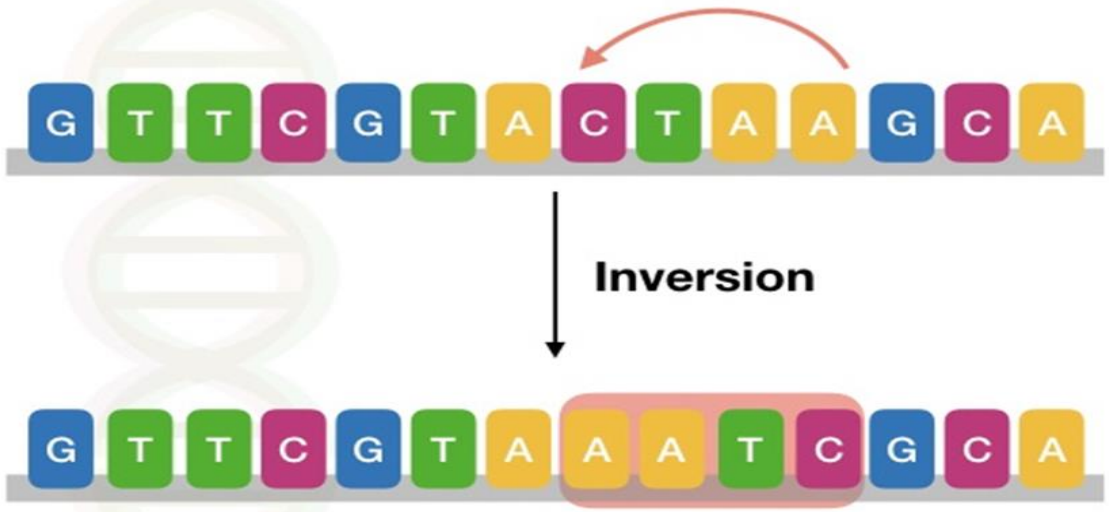
Modification of Mendelian ratios

Gene Mutations

- Types of gene mutations

1. **Substitution**- necleotide is replaced by another. The nucleotides work in triples and they code for a particular amino acid. When one nucleotide is changed the amino acid changes and it affects its function.
2. **Insertion/addition**- nucleotide(s) are added to a DNA strand
3. **Inversion**- nucleotides are cut off and pasted to the same strand in an inverted way
4. **Deletion**- the nucleotide(s) is cut and disappear


Substitution
Insertion
Deletion
Original sequence
TGGCAG
TGGCAG
TGGCAG
Mutated sequence
TGGTAG
TGGTATCAG
TGGG

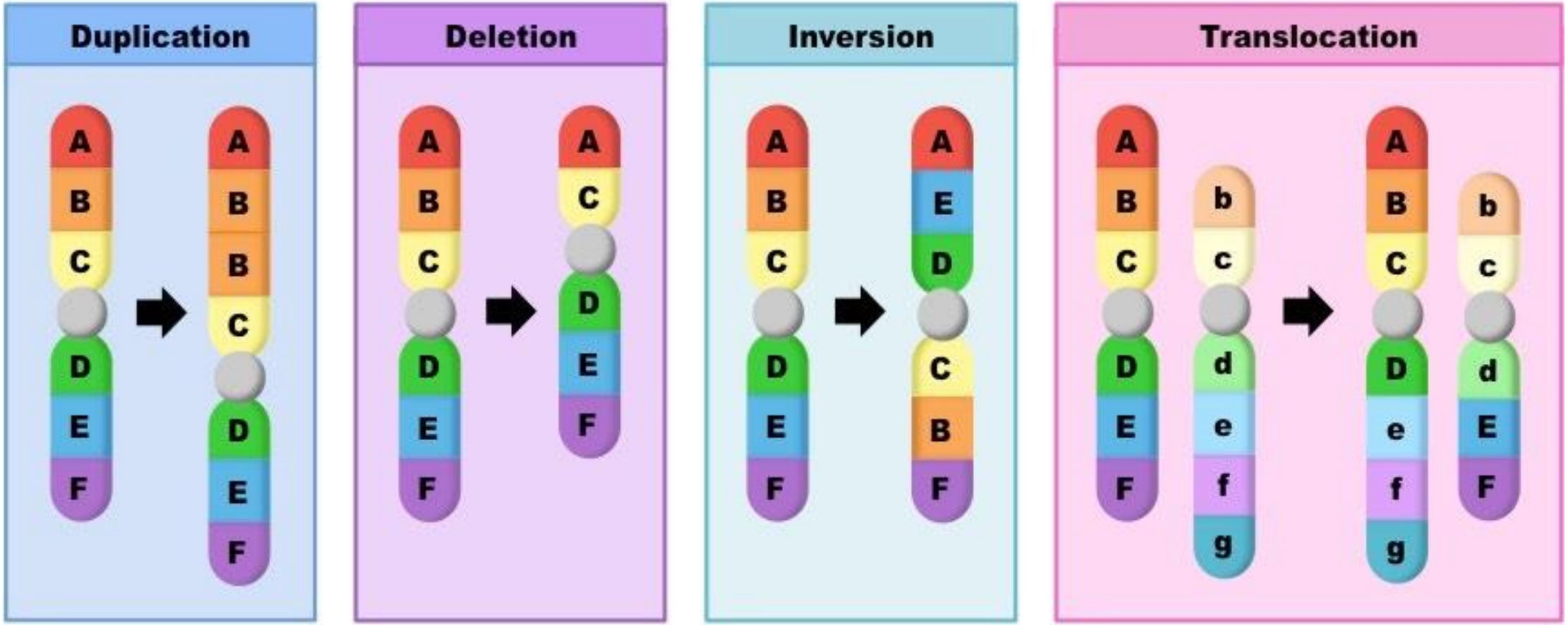

G T T C G T A C T A A G C A
Inversion
G T T C G T A A A T C G C A

Modification of Mendelian ratios

Chromosome mutations;

1. **Substitution**- a small part of the chromosome is cut and lost but is replaced by another part
2. **Translocation/ addition-** this involves transfer of a chromosome part to another chromosome
3. **Inversion**- part of a chromosome is cut and joins the same chromosome after flipping over.
4. **Deletion**- when part of a chromosome disappears
5. **Duplication** – a section of a chromosome is repeated

Chromosome mutations



Modification of Mendelian ratios

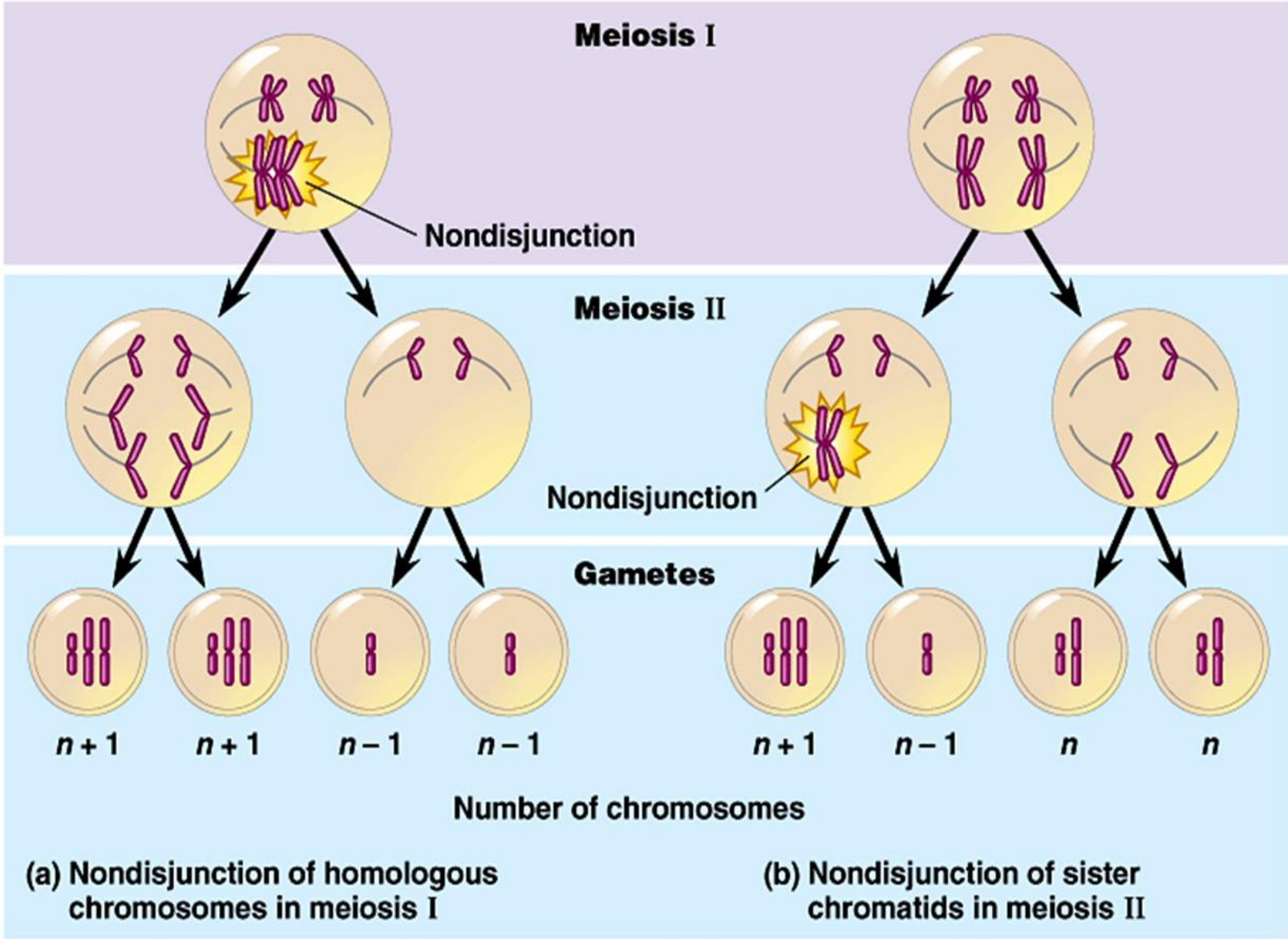
Chromosome mutations;

6. **Mutations involving a change in chromosome number-**

- Non-disjunction at anaphase of mitosis or meiosis may result in one cell having less or more than 2n which is called aneuploidy
- Monosomics have only one of a pair of a particular set of chromosomes, and trisomics have three copies of a chromosome, rather than the normal two

e.g. Male XY female XX .

Non disjunction in male may result in XO = tuner female, XXY = super male or XXX = super female



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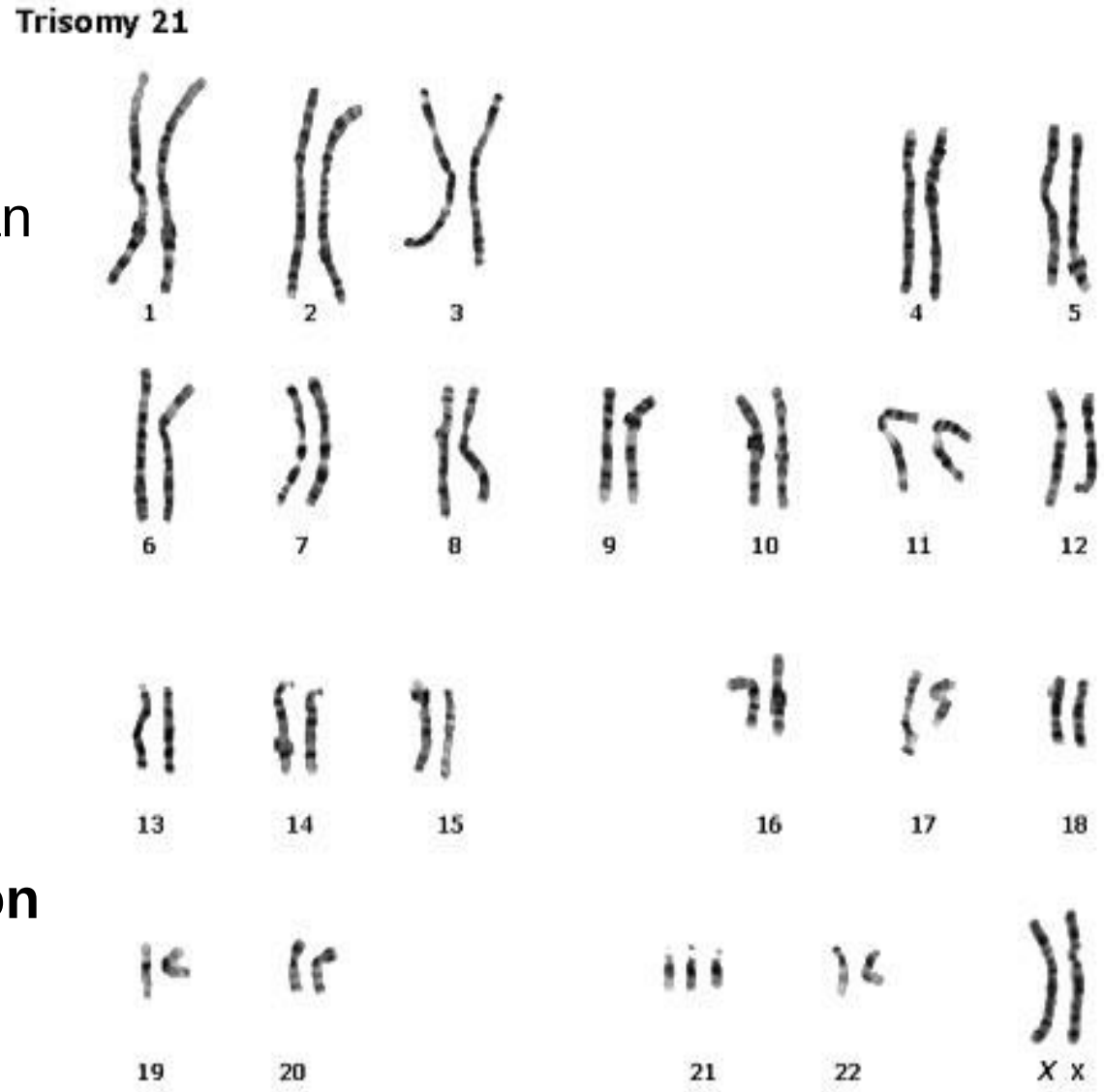
Non-disjunction Involving Autosomes

- Down syndrome is an example of a trisomic condition in which the individual is born with an extra copy of chromosome 21.
- This condition results in a small oval head, lower than normal IQ, short stature, reduced life span, and infertility in males.
- **Read on other disorders on non-disjunction on autosomes**



Nondisjunction Involving Sex Chromosomes

•Nondisjunction can cause individuals to be born female with one X (Turner syndrome), female with three X chromosomes (Trisomy X), male with XXY (Klinefelter syndrome), or male with XYY (XYY syndrome).